

FAST, MODULAR AND EXPLAINABLE GENERATIVE IMAGE RESTORATION

A DIFFUSION MODEL DISTILLATION & LANGEVIN SAMPLING APPROACH



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OUTLINE

01

BACKGROUND

02

LATINO-PRO: LATent consisTency INverse sOlver with PRompt Optimization

03

Learning few-step posterior samplers by unfolding and distillation

04

Conclusion

BACKGROUND

Problem statement:

Image data not useful in raw form (limited resolution, noise, blur..)



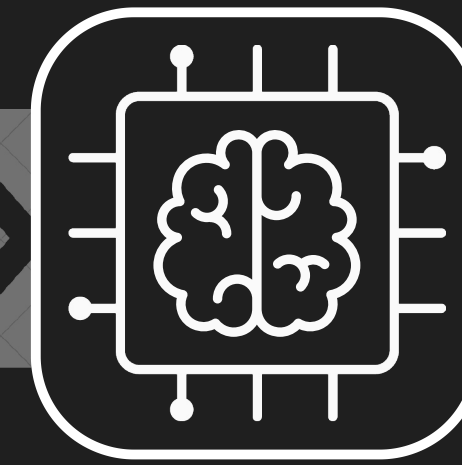
Unknown Image



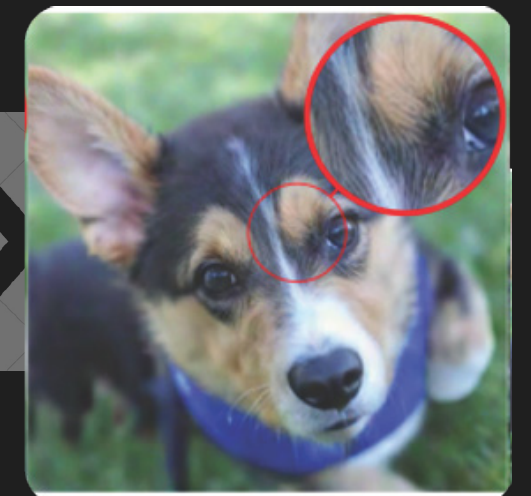
Instrument (limited resolution & noise)



Sensor Data



Computational Imaging



Recovered Image

Research vision:

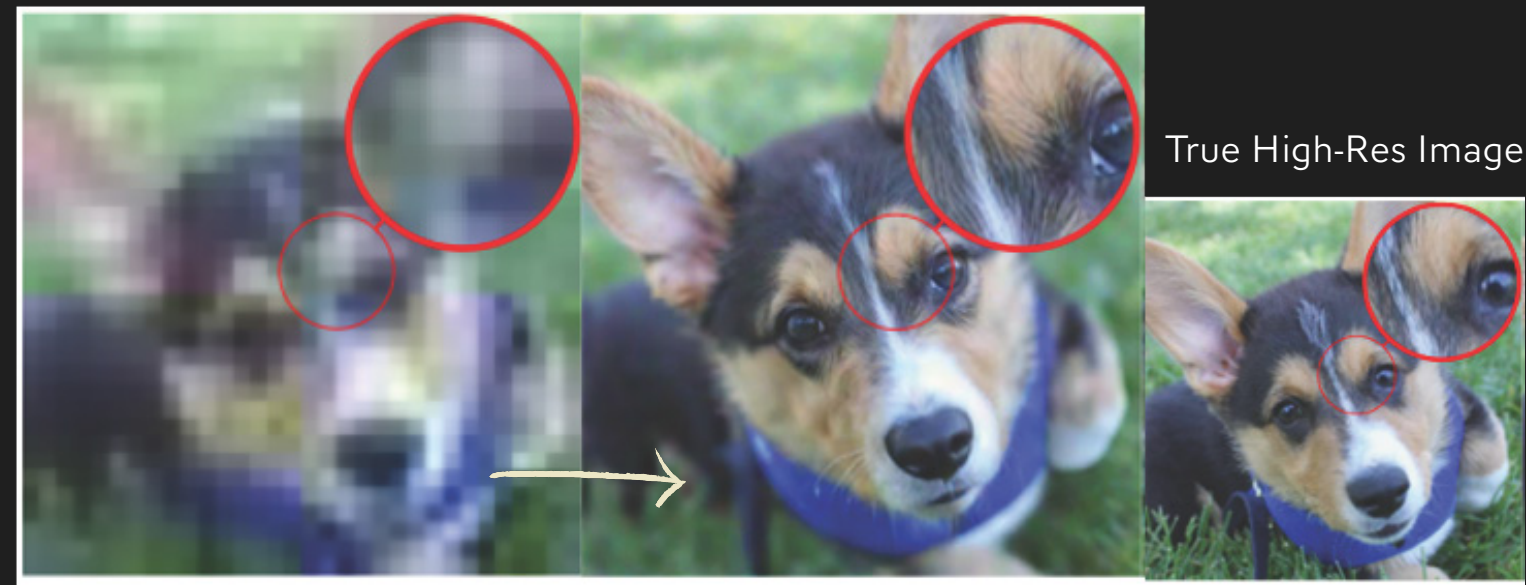
Upgrade imaging instruments through integrated physical and gen AI models, Bayesian statistics and fast stochastic algorithms.

TODAY'S TALK

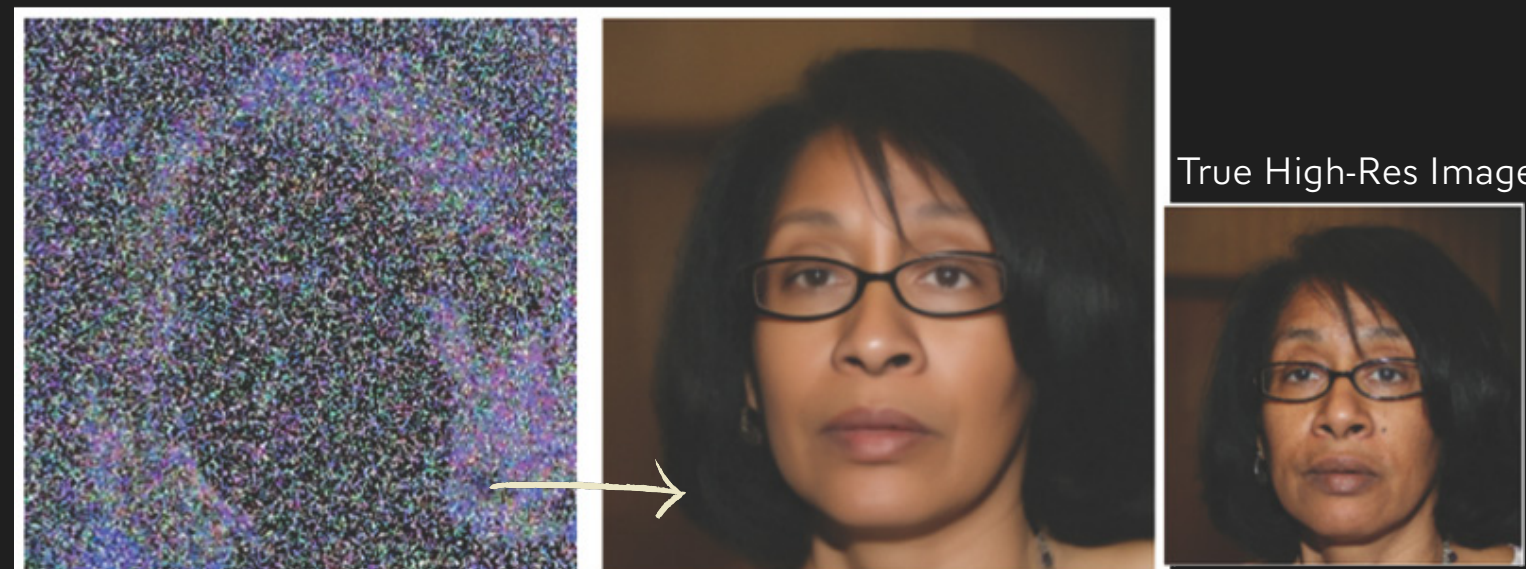
How modifying the mathematics of VLMs allows prompting w. physical model & measurement, while self-adjusting text prompt, transparently and with few NFEs.



Sample from text-image gen AI model
(Midjourney.com) prompted by text.



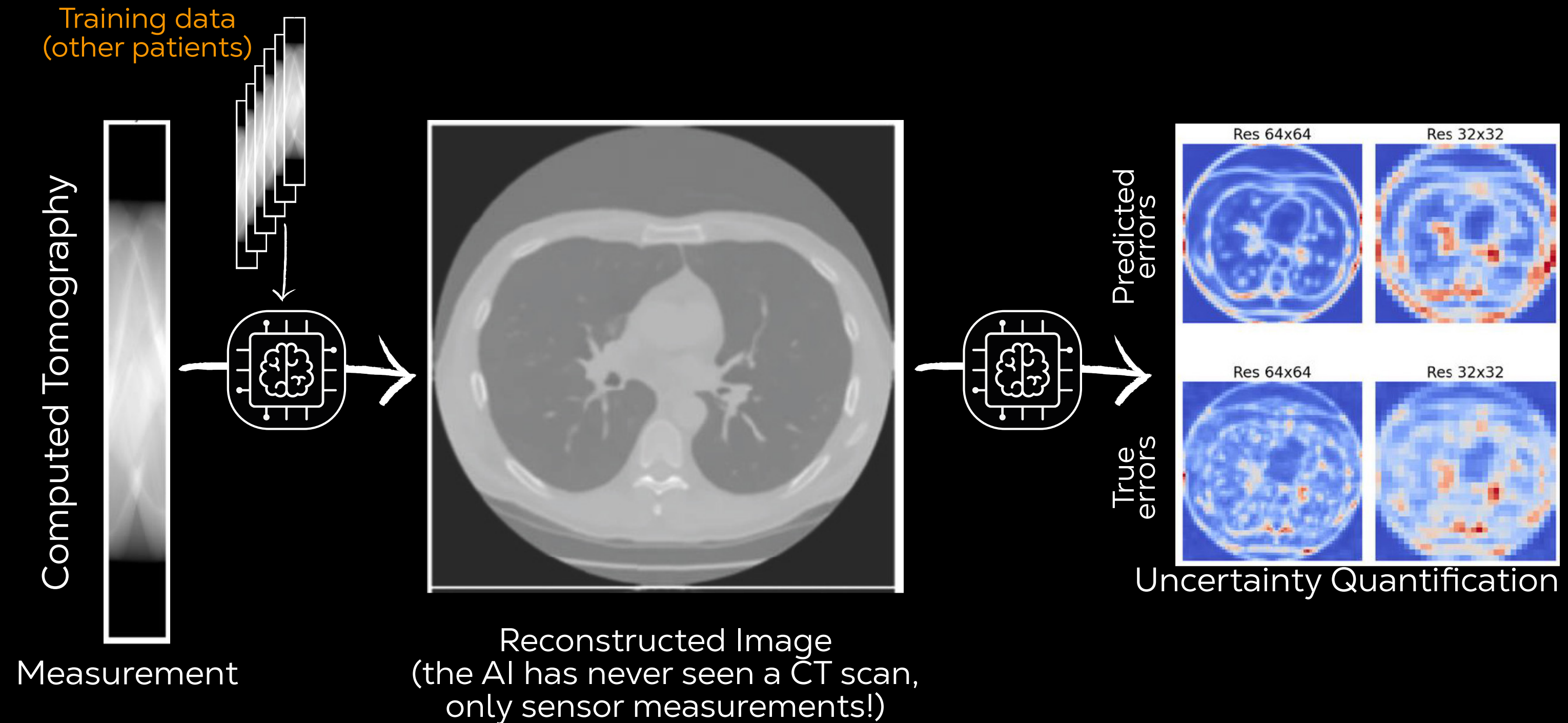
True High-Res Image



True High-Res Image

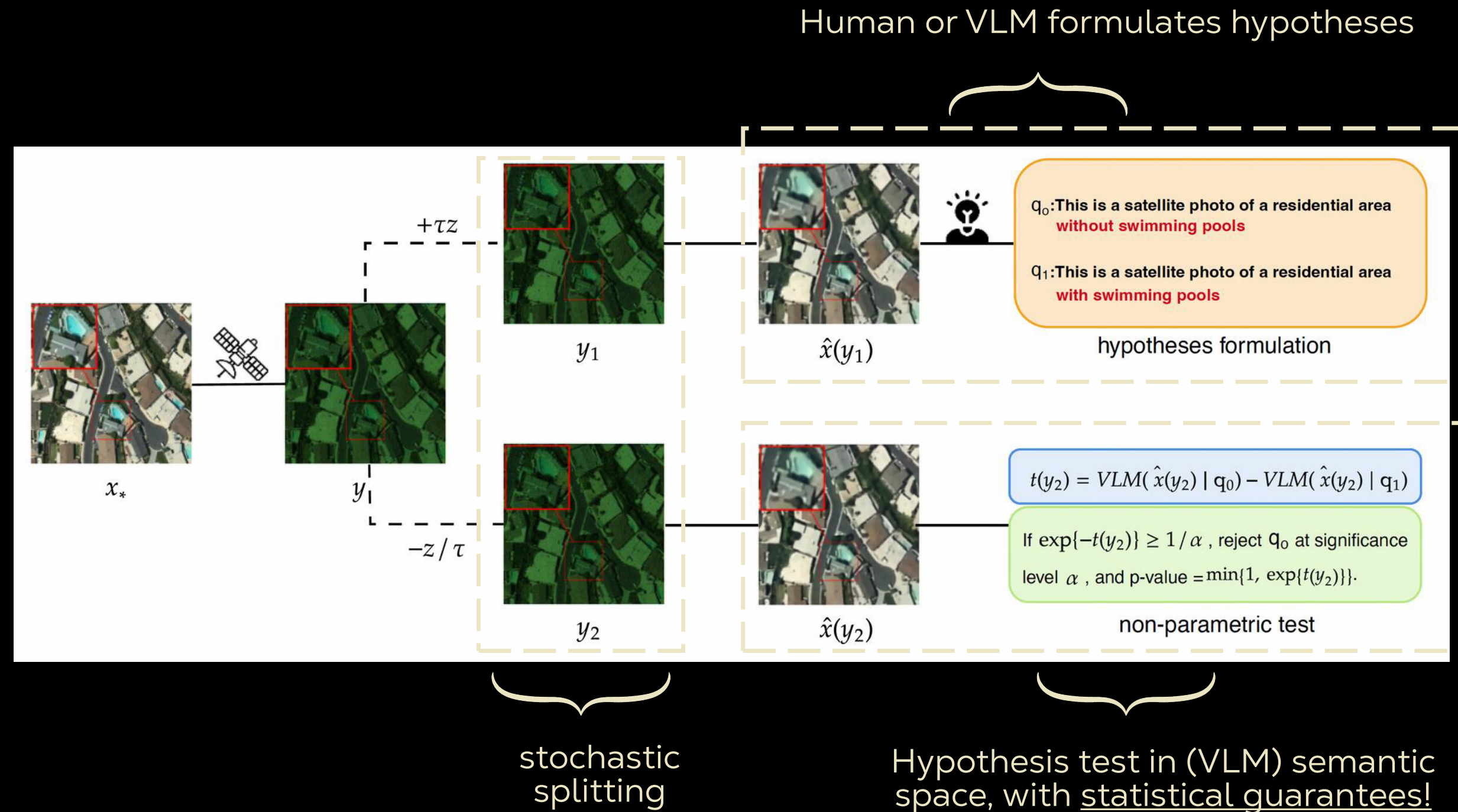
NOT TODAY

Self-supervised Gen-AI-based imaging: Leverage symmetries and invariance properties to learn empirical Bayesian models directly from the measurements – **no reference/clean/labelled data required!**

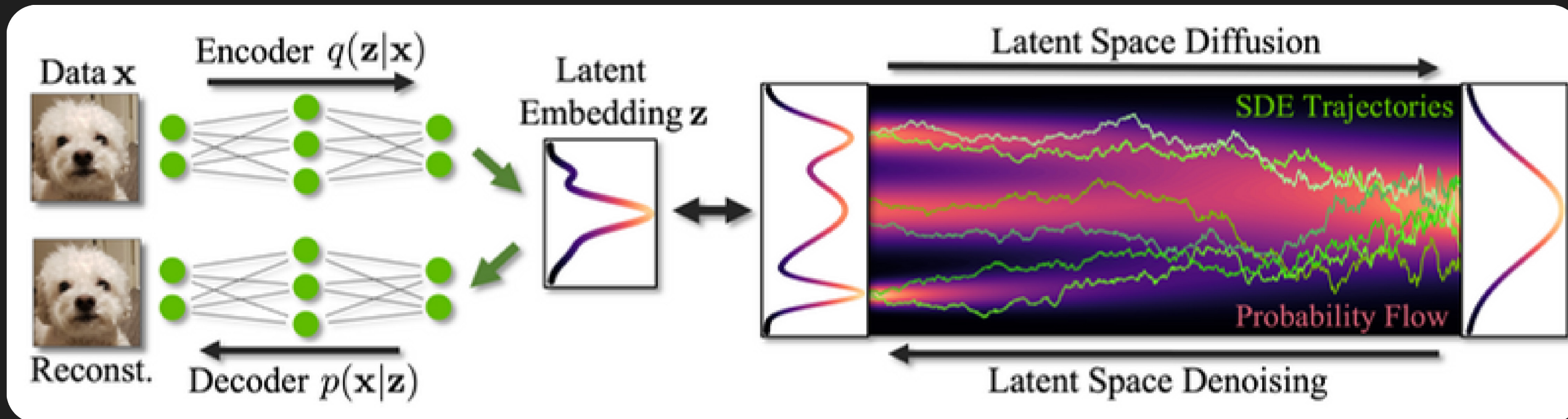


NOT TODAY

Hypothesis testing in imaging: Combine VLMs, *data fission*, and non-parametric statistics to probe semantic uncertainty & robustly test semantic hypotheses expressed in natural language.



LATENT DIFFUSION MODELS



Kreis et al. NeurIPS
2023, LDMS Tutorial

$$d\mathbf{z}_t = -\frac{\beta_t}{2}\mathbf{z}_t dt + \sqrt{\beta_t}d\mathbf{w},$$

$$d\mathbf{z}_t = \left[-\frac{\beta_t}{2}\mathbf{z}_t - \beta_t \nabla_{\mathbf{z}_t} \log p_t(\mathbf{z}_t) \right] dt + \sqrt{\beta_t}d\mathbf{w},$$

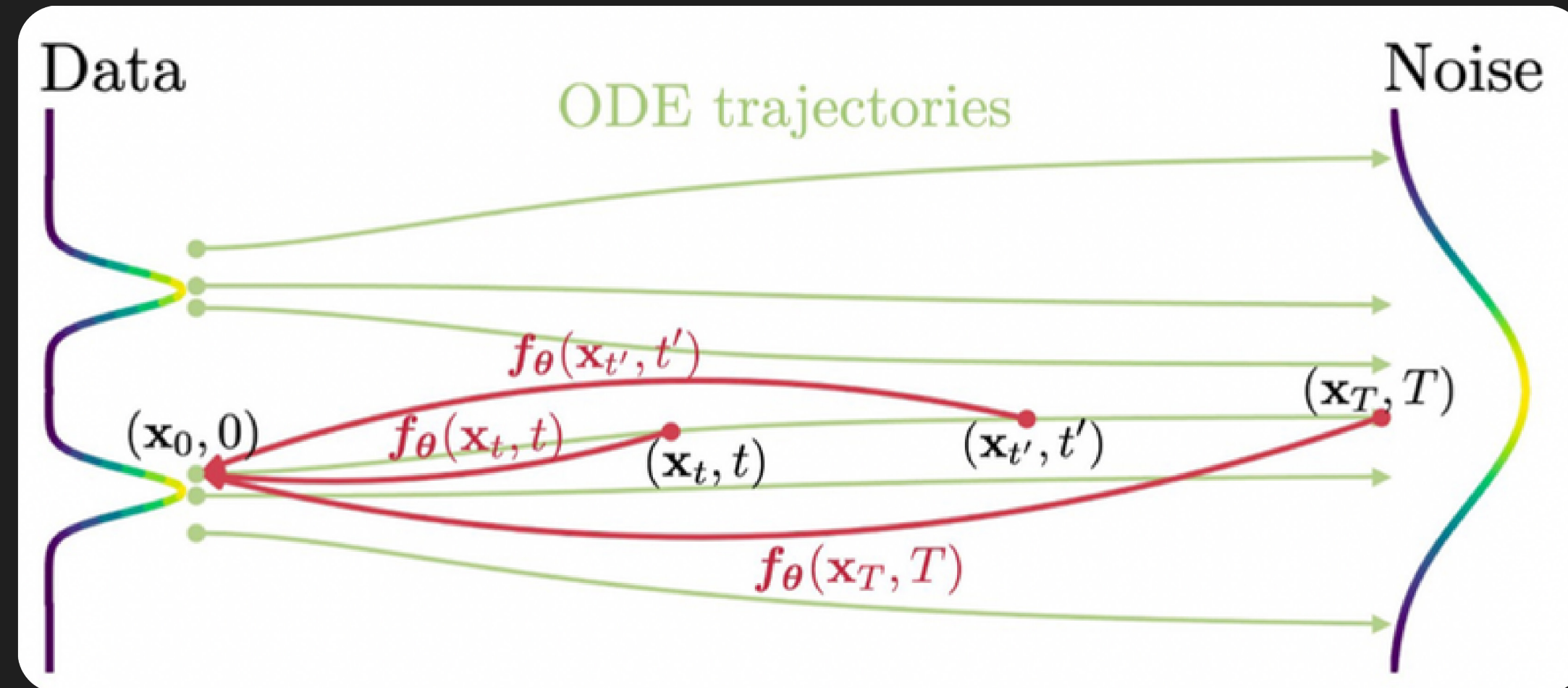
$$\mathcal{E} : \mathbb{R}^n \mapsto \mathbb{R}^d, \quad \mathcal{D} : \mathbb{R}^d \mapsto \mathbb{R}^n, \quad \mathbf{x} \approx \mathcal{D}(\mathcal{E}(\mathbf{x})),$$



arXiv:2405.14867

ACCELERATION

Probability Flow ODE, Distillation & Consistency Models (CMs)



arXiv:2303.01469

CMs are distilled diffusion models trained to transport any point on the ODE trajectory back to time 0. They are fast one-step samplers.

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03 Learning few-step posterior samplers by unfolding and distillation

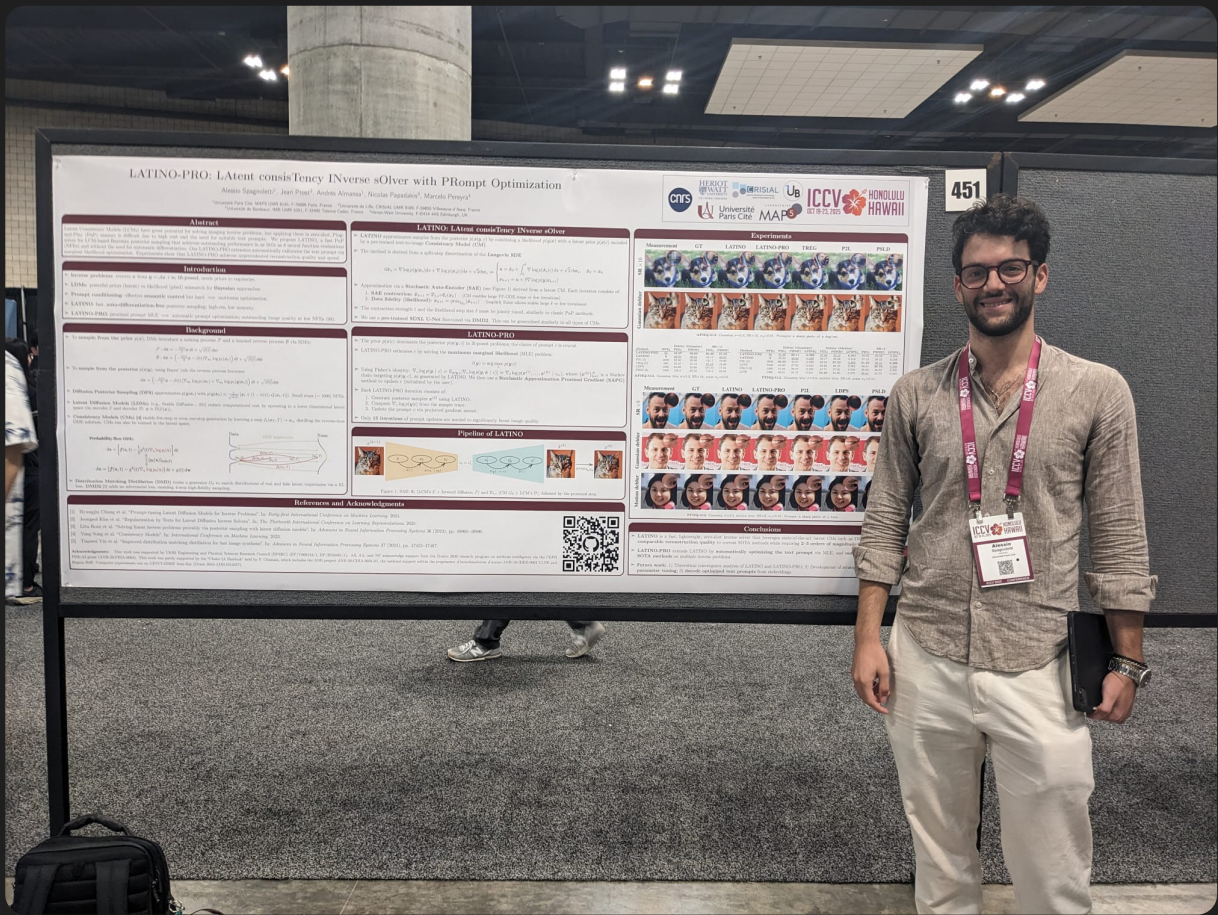
04 Conclusion

LATINO-PRO

Latent consistency INverse solver with PRompt Optimization

ICCV 2025

Joint work with Alessio Spagnoletti, Jean Prost, Andrés Almansa, and Nicolas Papadakis.



BAYESIAN STATISTICAL FRAMEWORK

We seek to perform inference on $x^* \in \mathbb{R}^d$

from some data $y = Ax^* + w$

We model x^* as a realisation of a r.v. $\mathbb{X}$

We model y as a realisation of a r.v. $(y|\mathbb{X} = x^*)$

We base our inferences about $\mathbb{X}$ on the posterior distribution

$$p(x|y) = \frac{p(y|x)p(x)}{\int_{\mathbb{R}^d} p(y|\tilde{x})p(\tilde{x})d\tilde{x}}$$

LANGEVIN SAMPLING

We sample the posterior by using the following diffusion process

$$d\mathbf{x}_s = \underbrace{\nabla \log p(\mathbf{y}|\mathbf{x}_s)}_{\text{data likelihood}} ds + \underbrace{\nabla \log p(\mathbf{x}_s|\mathbf{c})}_{\text{image prior}} ds + \sqrt{2}d\mathbf{w}_s$$

text prompt
↙

Converges exponentially fast to $p(\mathbf{x}|\mathbf{y})$ as time s increases.

Modular and explainable - clear data fidelity and regularization terms.

The *drift* is time-homogeneous, no need to approximate likelihoods.

Challenge: how to embed VLM priors (eg SDXL) within a Langevin process?

CORE IDEA 1

We propose to discretize the Langevin process as follows

$$\left. \begin{aligned} \boldsymbol{u} &= \boldsymbol{x}_k + \int_0^\delta \nabla \log p(\tilde{\boldsymbol{x}}_s | c) \mathrm{d}s + \sqrt{2} \mathrm{d}\boldsymbol{w}_s, \tilde{\boldsymbol{x}}_0 = \boldsymbol{x}_k, \\ \boldsymbol{x}_{k+1} &= \boldsymbol{u} + \delta \nabla \log p(\boldsymbol{y} | \boldsymbol{x}_{k+1}), \end{aligned} \right\} \begin{array}{l} \text{image prior} \\ \text{data likelihood} \end{array}$$

The top line corresponds to a Langevin process targeting the prior.

The bottom line is an implicit or *proximal* step, exactly solvable.

Insight: can replace top line with other stochastic steps that contract random variables towards the prior.

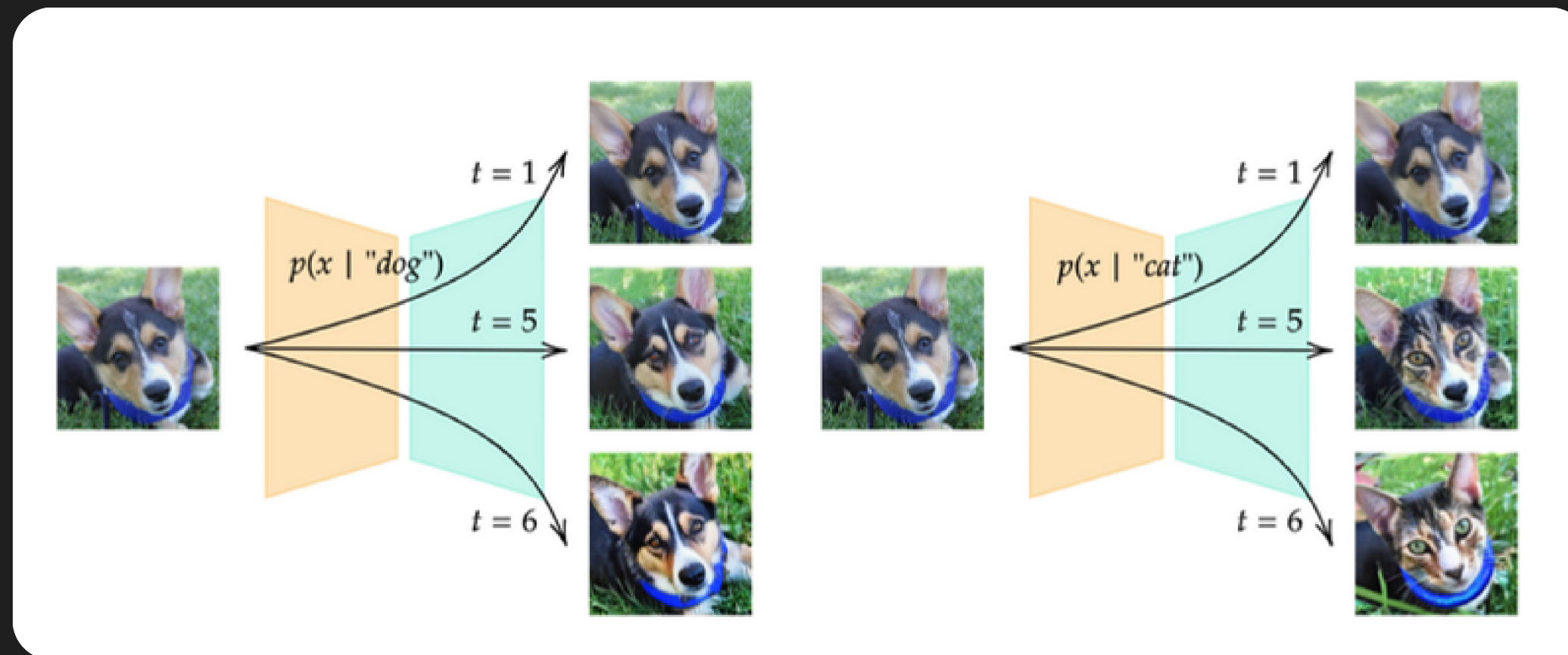
CORE IDEA 2

Auto-encode (distilled) DMs to contract random variables towards their internal generative model

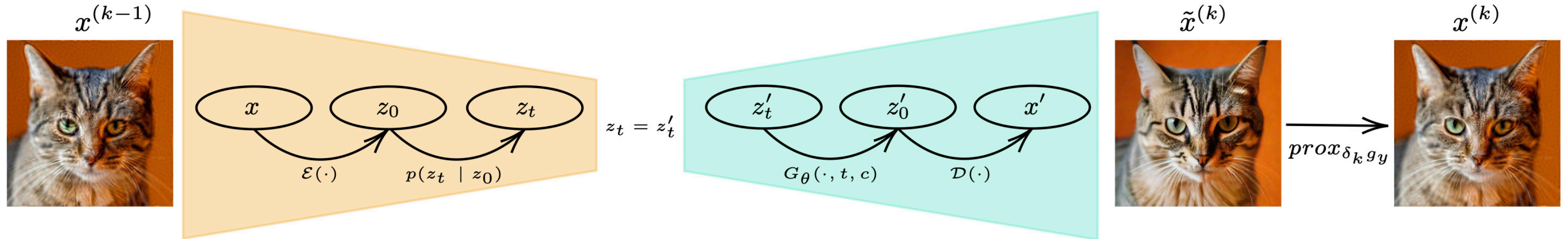
$$\mathfrak{E}_t : z_t | \mathbf{x} \sim \mathcal{N}(\sqrt{\alpha_t} \mathcal{E}(\mathbf{x}), (1 - \alpha_t) \text{Id}_d)$$

$$\mathfrak{D}_{t,c} : \mathbf{x}' = \mathcal{D}(G_\theta(z'_t, t, c))$$

generative
model



LATINO



```

for  $k = 1, \dots, N$  do
   $\epsilon \sim \mathcal{N}(0, \text{Id})$ 
   $z_{t_k}^{(k)} \leftarrow \sqrt{\alpha_{t_k}} \mathcal{E}(x^{(k-1)}) + \sqrt{1 - \alpha_{t_k}} \epsilon$   $\triangleright$  Encode
   $u^{(k)} \leftarrow \mathcal{D}(G_\theta(z_{t_k}^{(k)}, t_k, c))$   $\triangleright$  Decode
   $x^{(k)} \leftarrow \text{prox}_{\delta_k g_y}(u^{(k)})$   $\triangleright g_y : x \mapsto -\log p(y|x)$ 
end for
  
```


CORE IDEA 3

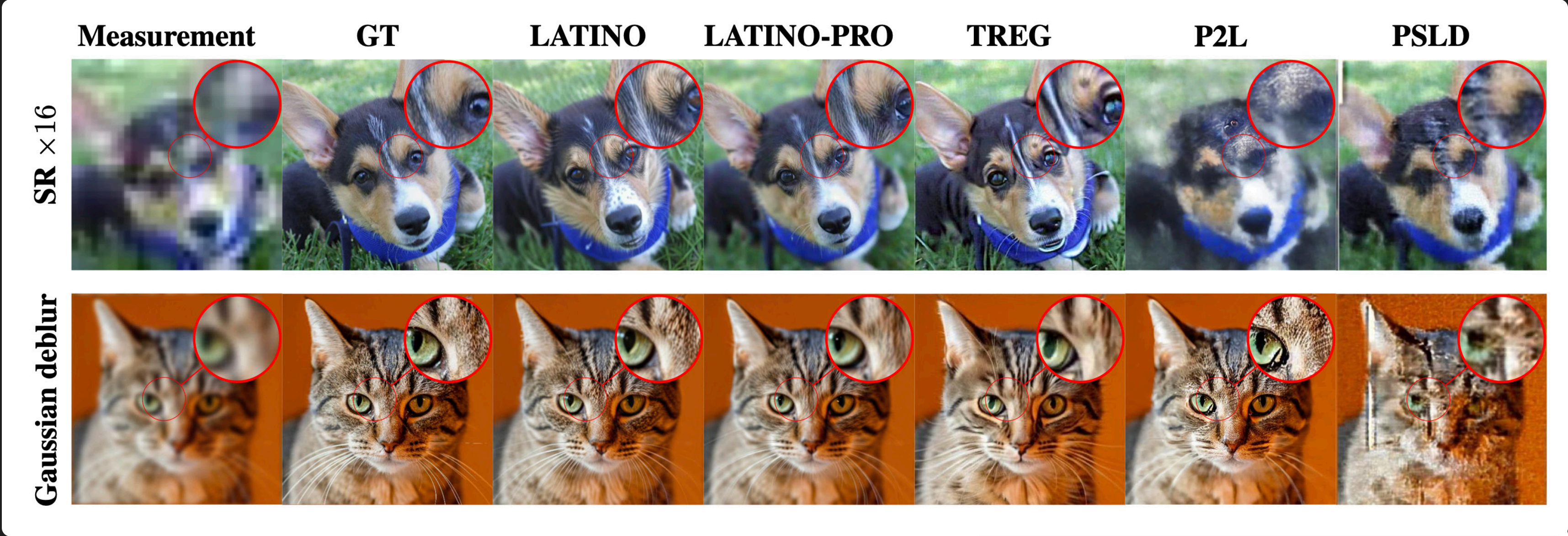
Self-calibrate text prompt by maximum marginal likelihood optimization

```
for  $m = 1, \dots, M$  do  
  for  $k = 1, \dots, N_m$  do ▷ LATINO  
     $\epsilon \sim \mathcal{N}(0, \text{Id})$   
     $\mathbf{z}_{t_k}^{(k)} \leftarrow \sqrt{\alpha_{t_k}} \mathcal{E}(\mathbf{x}^{(k-1)}) + \sqrt{1 - \alpha_{t_k}} \epsilon$   
     $\mathbf{u}^{(k)} \leftarrow \mathcal{D}(G_\theta(\mathbf{z}_{t_k}^{(k)}, t_k, c_m))$   
     $\mathbf{x}^{(k)} \leftarrow \text{prox}_{\delta_k g_y}(\mathbf{u}^{(k)})$   
  end for  
   $h(c_m) \leftarrow \nabla_c \log p(\mathbf{z}_{t_1}^{(1)}, \dots, \mathbf{z}_{t_{N_m}}^{(N_m)} | c_m)$   
   $c_{m+1} = \Pi_C [c_m + \gamma_m h(c_m)]$  ▷ SAPG  
   $\mathbf{x}^{(0)} \leftarrow \mathbf{x}^{(N_m)}$  ▷ Carry state forward
```

} stochastic approx. proximal
gradient step to compute

$$\hat{c}(\mathbf{y}) = \arg \max_{c \in \mathbb{R}^k} p(\mathbf{y} | c)$$

SOME RESULTS



P2L: Chung et al. ICML 2024, TREG: Kim et al. ICLR 2025, PSLD: Rout et al. NeurIPS 2023.

M PEREYRA

Method	NFE↓	Deblur (Gaussian)		SR $\times 16$	
		FID↓	PSNR↑	FID↓	PSNR↑
LATINO-PRO	<u>68</u>	18.37	26.82	30.40	21.52
LATINO	8	<u>20.03</u>	<u>26.25</u>	42.14	<u>20.05</u>
P2L [7]	2000	85.80	20.96	121.7	19.99
TReg [21]	200	35.47	21.13	<u>37.13</u>	19.60
LDPS	1000	64.88	22.60	101.13	17.34
PSLD [45]	1000	125.5	20.52	113.4	16.48

SOME RESULTS

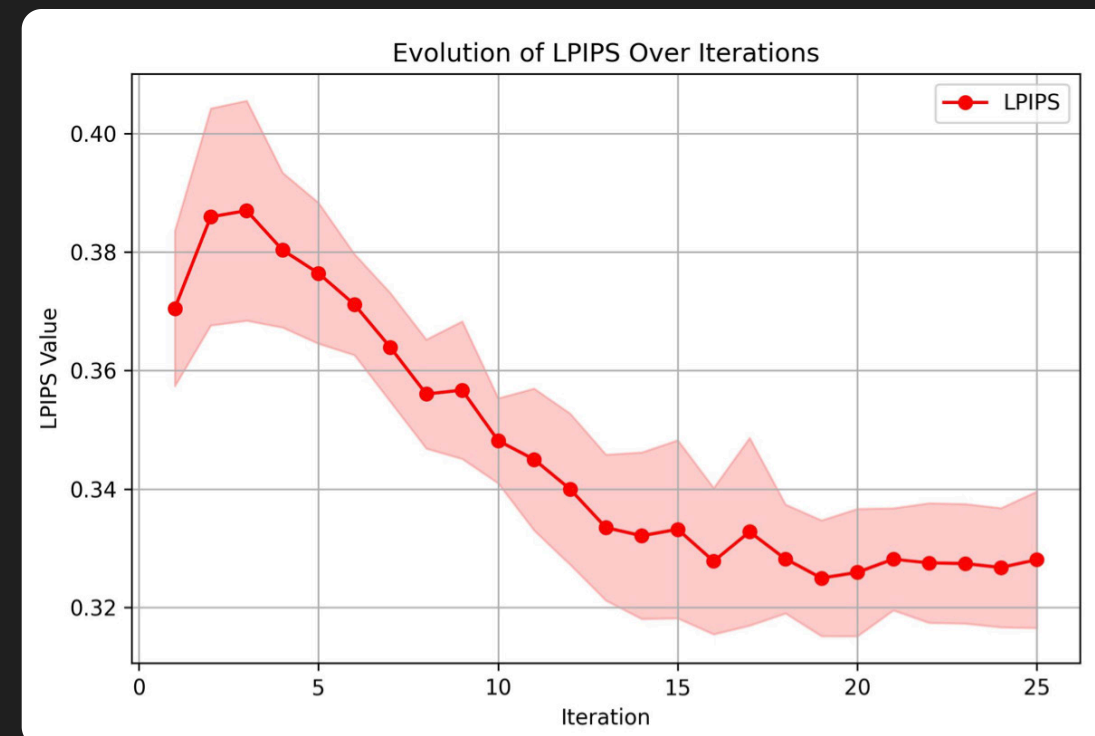
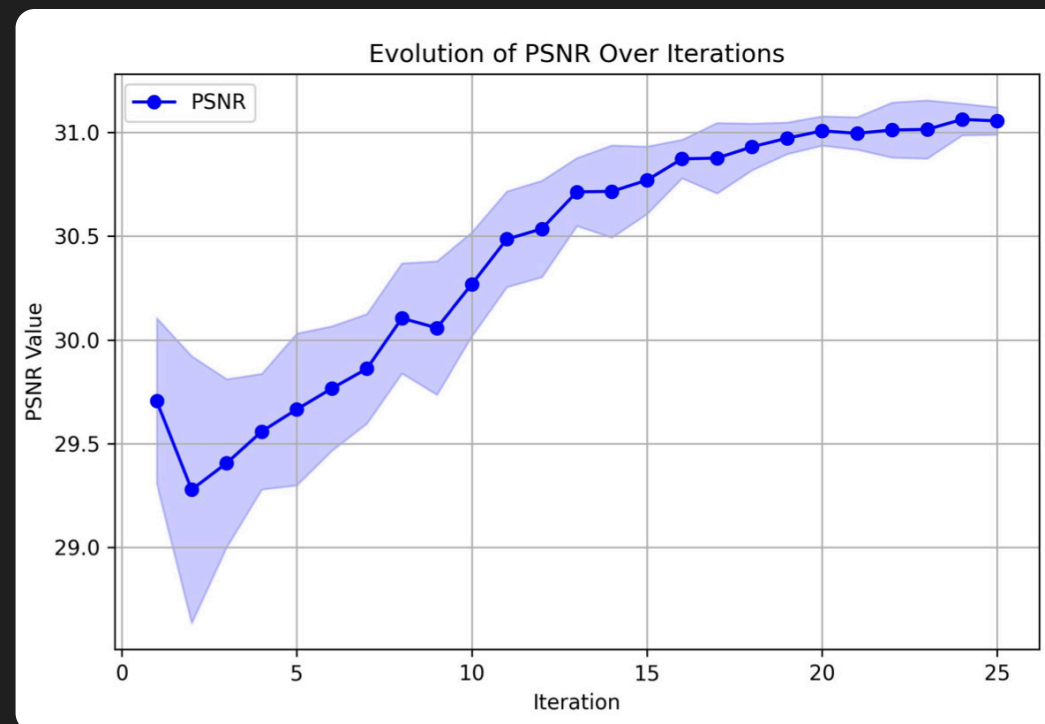


Method	Deblur (Gaussian)				Deblur (Motion)			SR×8		
	NFE↓	FID↓	PSNR↑	LPIPS↓	FID↓	PSNR↑	LPIPS↓	FID↓	PSNR↑	LPIPS↓
LATINO-PRO	68	31.98	29.11	0.292	27.80	27.14	0.301	40.95	26.58	0.355
LATINO	8	33.94	28.95	0.296	29.17	26.88	0.318	37.13	26.22	0.356
P2L [7]	2000	30.62	26.97	0.299	28.34	27.23	<u>0.302</u>	31.23	<u>28.55</u>	0.290
LDPS	1000	45.89	27.82	0.334	58.66	26.19	0.382	36.81	28.78	<u>0.292</u>
PSLD [45]	1000	41.04	28.47	0.320	47.71	27.05	0.348	36.93	26.62	0.335

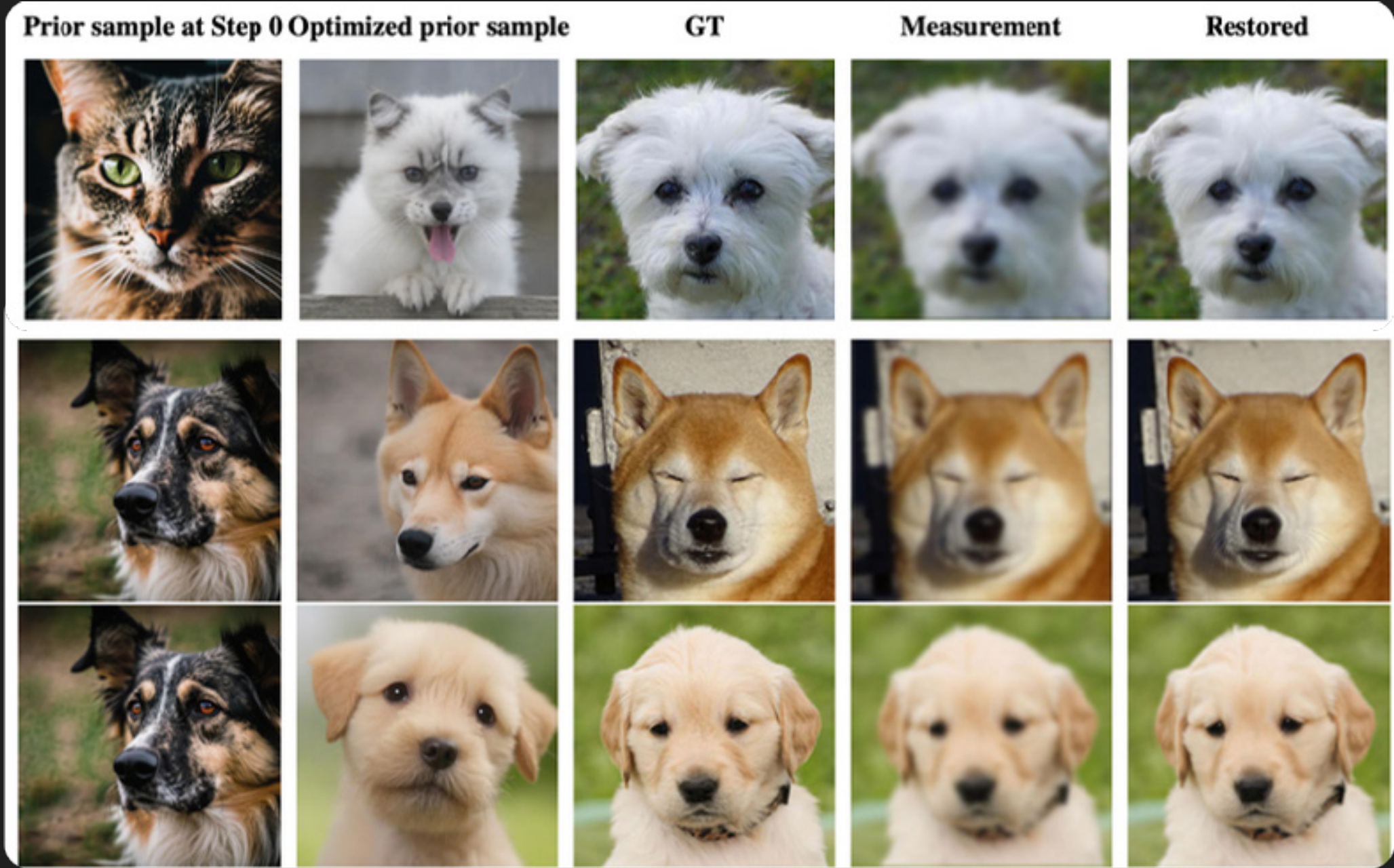
VISUALIZATION OF PROMPT OPTIMIZATION



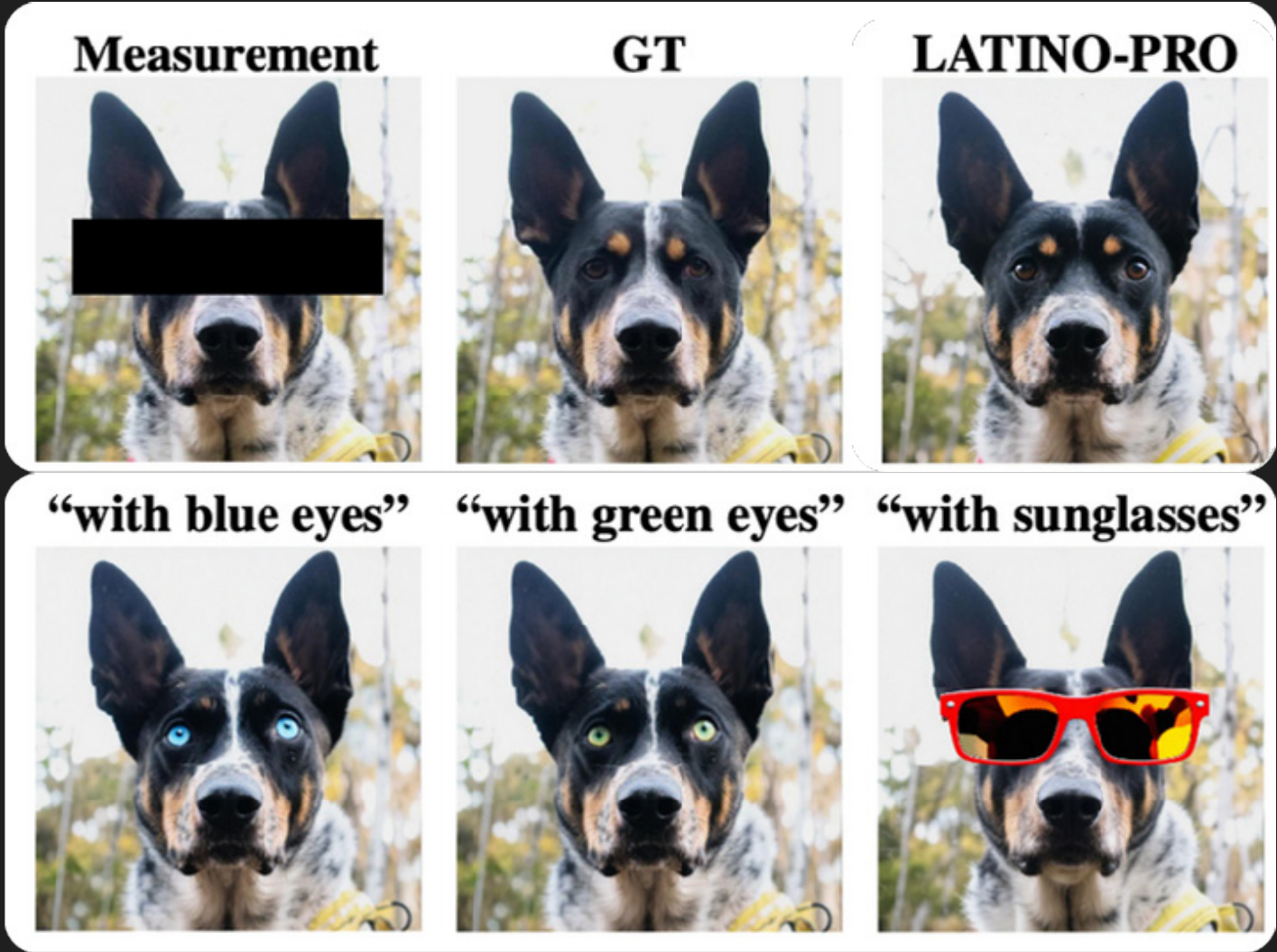
Samples from the prior before/after 4 SAPG steps



VISUALIZATION OF PROMPT OPTIMIZATION

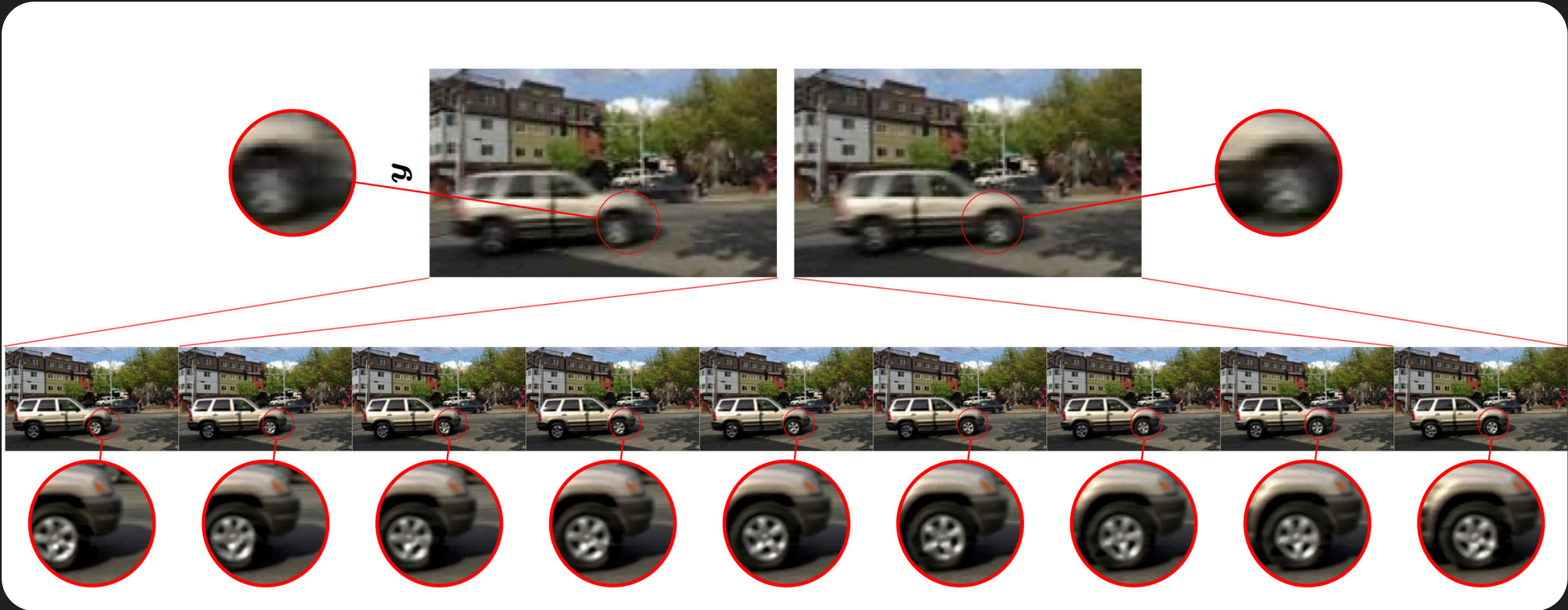


Samples from the prior before/after 4 SAPG steps



Samples with constrained SAPG steps to enforce semantic constraints

EXTENSION TO VIDEO RESTORATION



Spatial-temporal super-resolution x8
using video CM distilled from WAN 2.1
(see <https://arxiv.org/pdf/2510.01339>)

Method	Problem C: Temp. SR×8 + SR×8				
	NFE↓	FVMD↓	PSNR↑	SSIM↑	LPIPS↓
LVTINO	7	602.5	23.11	0.697	0.411
VISION-XL	8	1604	23.38	0.652	0.520
ADMM-TV	–	1645	18.15	0.663	0.439

EXTENSION TO VIDEO RESTORATION

Spatial-temporal super-resolution x8 as a challenging linear problem



EXTENSION TO VIDEO NON-LINEAR RESTORATION

We use JPEG artefact removal as non-linear test case



EXTENSION TO VIDEO NON-LINEAR RESTORATION

We use JPEG artefact removal as non-linear test case



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Learning few-step posterior samplers by unfolding and distillation of diffusion models

TMLR 2025.

Joint work with Charlesquin Kemajou Mbakam and Jonny Spence.

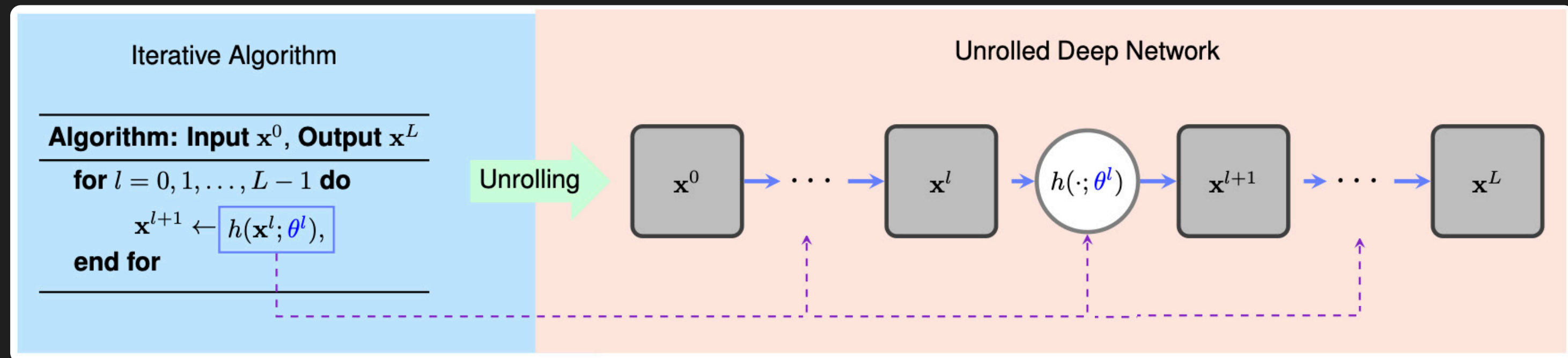


BACKGROUND

Deep unfolding:

Transforms an iterative (optimization) algorithm with a fixed number of iterations into a deep neural network architecture, whereby the algorithm's steps become trainable layers that are trained end-to-end.

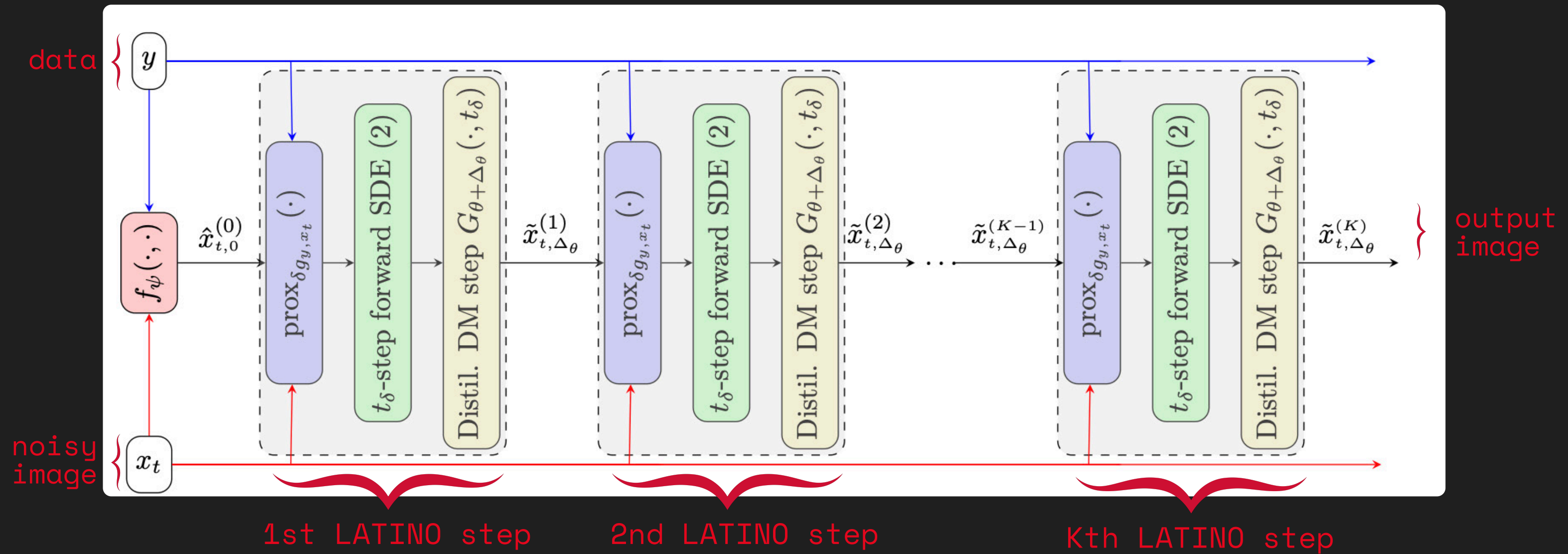
A powerful template for designing interpretable architectures that incorporate y and the degradation model explicitly during inference.



arXiv:1912.10557

CORE IDEA 1

We propose to unfold K LATINO modules. As prior, we use a DM $G_{\theta+\Delta_\theta}$ where Δ_θ is a *LoRA* on a frozen pre-trained DM G_θ . Warm-start with f .



CORE IDEA 2

We distil as conditional CM to sample $\mathbb{x}$ given y and a noisy copy of $\mathbb{x}$:

$$\begin{aligned} \arg \min_{\Delta_{\theta}, \psi} \quad & \mathcal{L}^G(\Delta_{\theta}, \psi) \triangleq \mathcal{L}_{\text{Adv}}(\Delta_{\theta}, \psi, \phi) + \omega_{\ell_2} \mathcal{L}_{\ell_2}(\Delta_{\theta}, \psi) + \omega_{\text{PS}} \mathcal{L}_{\text{PS}}(\Delta_{\theta}, \psi), \\ \arg \max_{\phi} \quad & \mathcal{L}^D(\phi) \triangleq \mathcal{L}_{\text{Adv}}(\Delta_{\theta}, \vartheta, \phi) + \omega_{\text{GS}} \mathcal{L}_{\text{GS}}(\phi), \end{aligned}$$

Conditional consistency model objective

$$\mathcal{L}_{\ell_2}(\Delta_{\theta}, \psi) = \mathbb{E}_{t, \mathbf{x}_t, \mathbf{y}, \mathbf{x}_0} \left[\|\mathbf{x}_0 - L_{\Delta_{\theta}, \psi}(\mathbf{x}_t, \mathbf{y})\|_2^2 \right]$$

MSE quality loss

$$\mathcal{L}_{\text{Adv}}(\Delta_{\theta}, \vartheta, \phi) = \mathbb{E}_{\mathbf{x}_0, \mathbf{y}} \left[\log(\varsigma(D_{\phi}(\mathbf{x}_0; \mathbf{y}))) \right] + \mathbb{E}_{t, \mathbf{x}_t, \mathbf{y}} \left[\log(1 - \varsigma(D_{\phi}(L_{\Delta_{\theta}, \psi}(\mathbf{x}_t, \mathbf{y}), \mathbf{y}))) \right]$$

adversarial (GAN) loss

$$\mathcal{L}_{\text{PS}}(\Delta_{\theta}, \psi) = \mathbb{E}_{t, \mathbf{x}_t, \mathbf{y}, \mathbf{x}_0} \left[\text{LPIPS}(\mathbf{x}_0, L_{\Delta_{\theta}, \psi}(\mathbf{x}_t, \mathbf{y})) \right]$$

perceptual quality loss

$$\mathcal{L}_{\text{GS}}(\phi) = \mathbb{E}_{t, \mathbf{x}_t, \mathbf{y}, \mathbf{x}_0, \mathbf{u}} \left[\left\| \nabla_x D_{\phi}(\mathbf{u}\mathbf{x} + (1 - \mathbf{u})L_{\Delta_{\theta}, \psi}(\mathbf{x}_t, \mathbf{y}), \mathbf{y}) \right\|^2 \right]$$

Discriminator regularity loss

UNFOLDED DISTILLED DIFFUSION MODEL (UD2M)

Embed trained UD2M network within multi-step CM sampler. The data y and the **data fidelity term are specified during inference.**

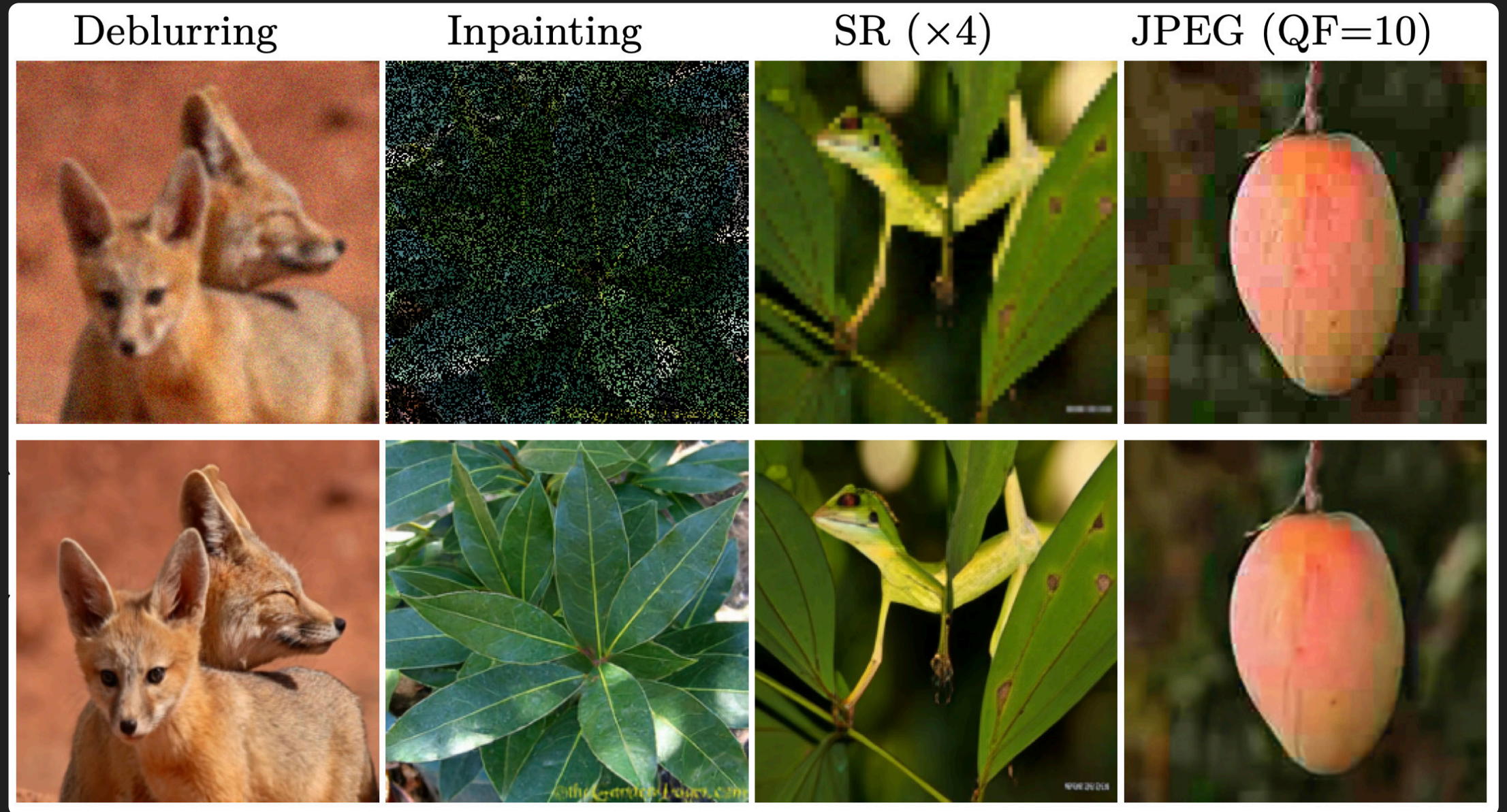
Require: Observation y , Time-grid $0 = t_0 < t_1 < \dots < t_N = T$,

- 1: Sample $x_{t_N} \sim \mathcal{N}(0, I)$ ▷ Initialize reversed diffusion
- 2: **for** $n = N, \dots, 1$ **do**
- 3: Set $\hat{x}_0 \leftarrow \tilde{x}_{t_n, K}^{\Delta\theta}(x_{t_n}, y)$ using L_ϑ (see I ▷ Unfolded sample targeting $p_0(x_0 | x_{t_n}, y)$
- 4: Sample $x_{t_{n-1}} \sim p_{t_{n-1}}(x_{t_{n-1}} | \hat{x}_0, x_{t_n})$ ▷ Reverse DDIM step
- 5: **end for**
- 6: **return** x_{t_0}

We typically unfold $K = 3$ LATINO modules with warm-starting and use $N = 3$ steps.

ILLUSTRATIVE EXAMPLES

UD2M samples on the ImageNet 256 dataset for Gaussian Deblurring, random inpainting (70%), super-resolution (4 $\times$), and restoration of JPEG compression artifacts (QF=10). All methods are implemented with an ImageNet DM in pixel domain (no text prompting).

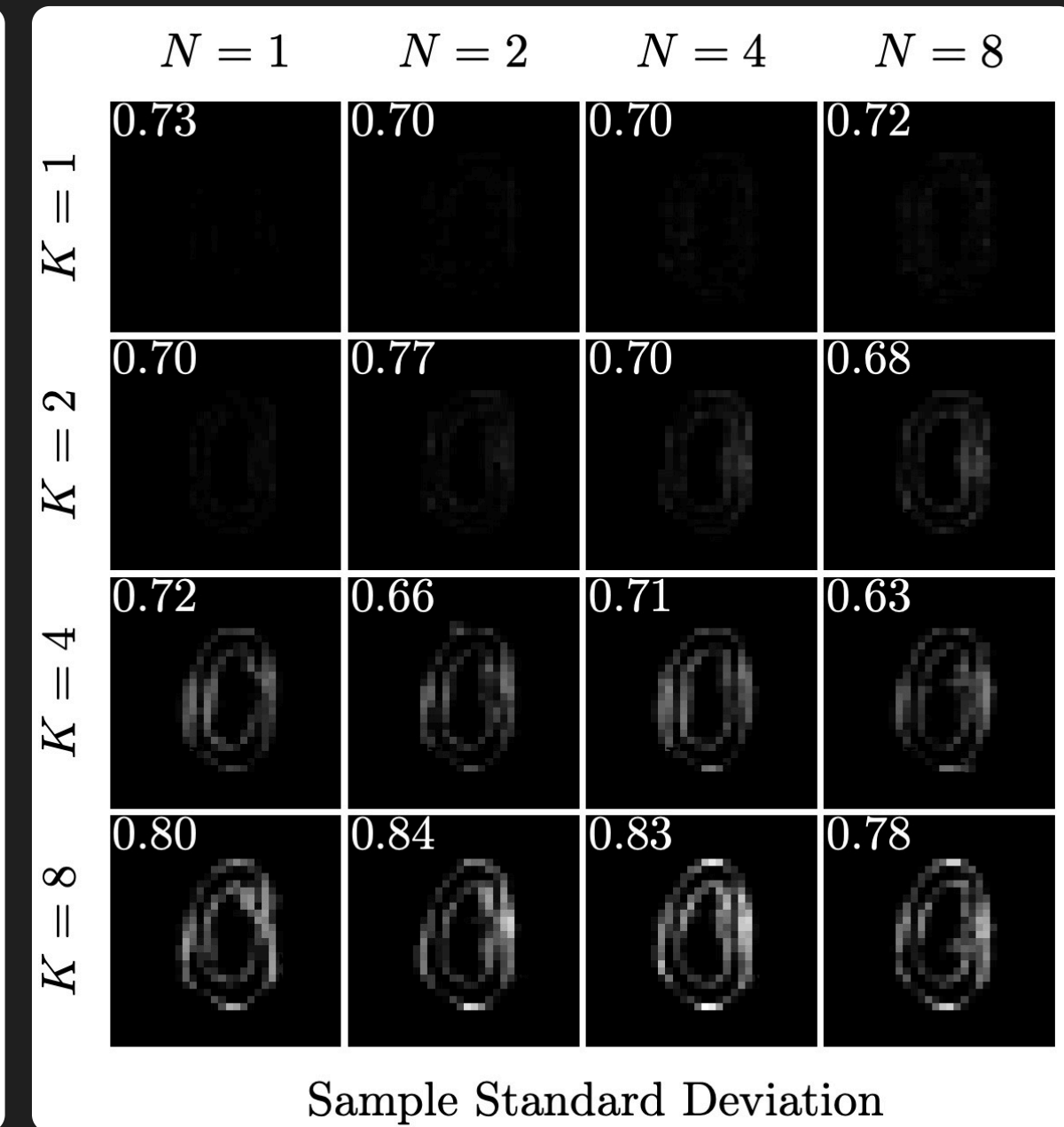
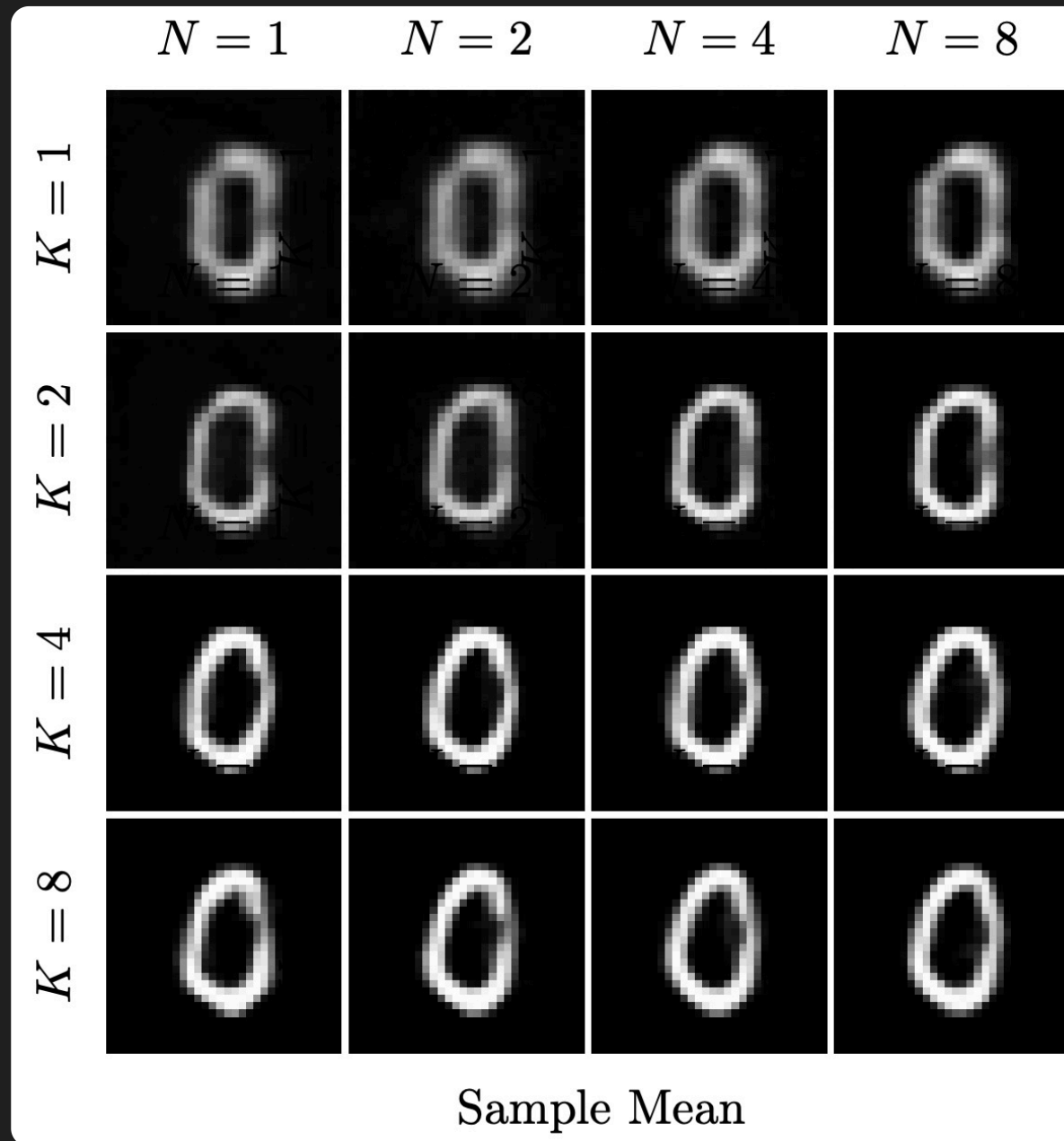
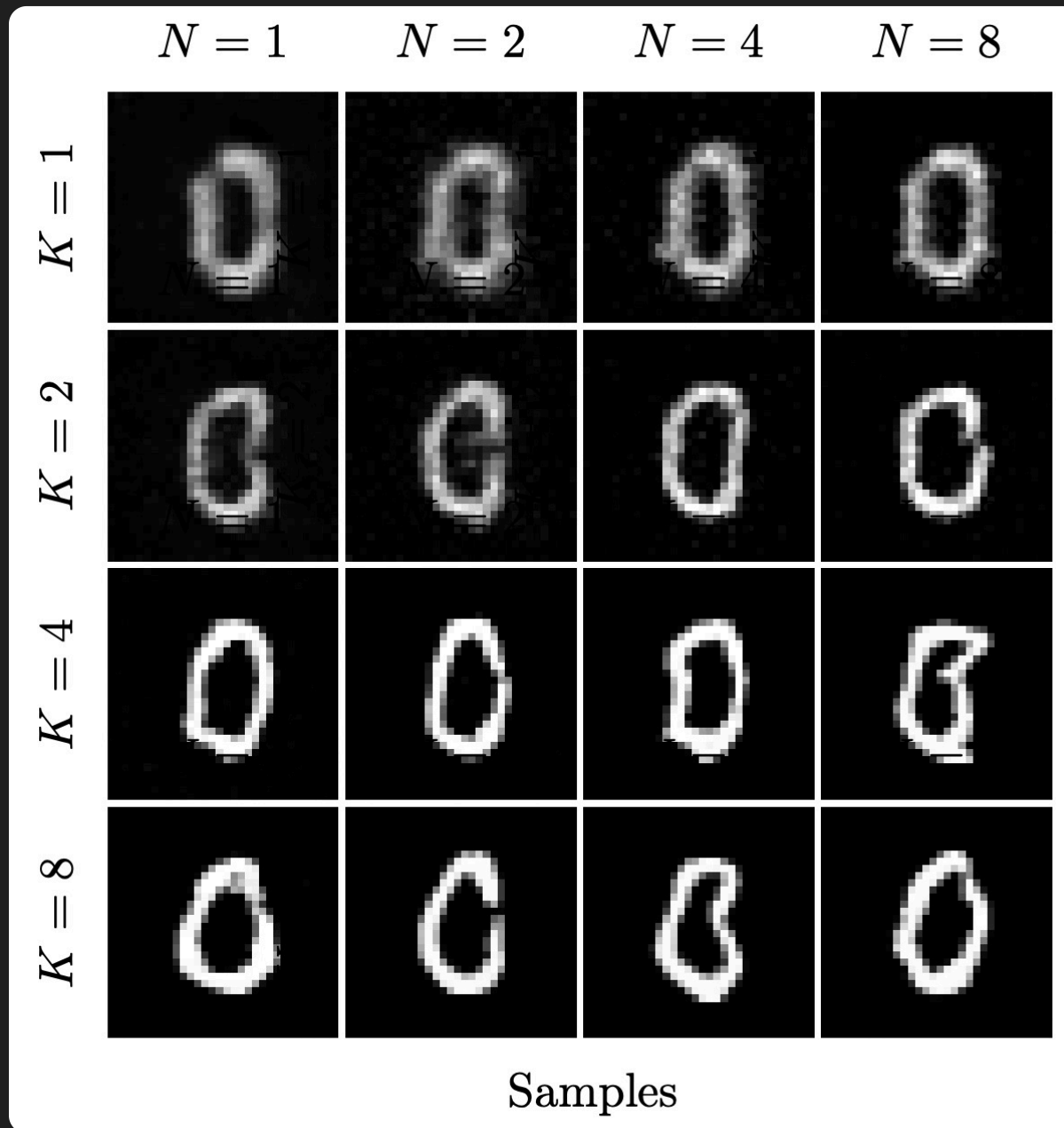
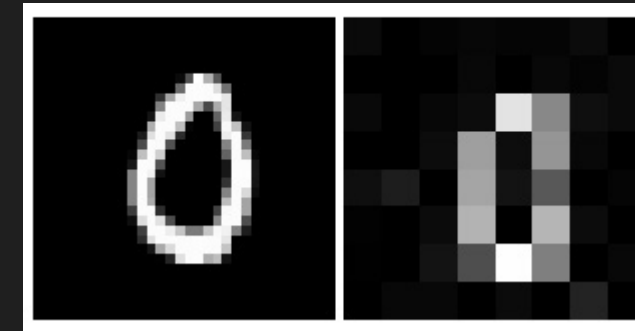


Methods	NFEs	Deblurring						SR $\times 4$			JPEG QF=10		
		Gaussian			Uniform			PSNR	LPIPS	FID	PSNR	LPIPS	FID
Ours (wo RAM)	9	38.77	0.02	4.61	35.57	0.02	11.14	24.42	0.15	20.69	27.52	0.18	35.16
Ours (w/ RAM)	12	35.97	0.01	3.30	36.96	0.01	2.69	26.70	0.08	11.9	-	-	-
CDDb	1000	37.02	0.06	5.01	31.26	0.19	23.15	26.41	0.2	19.88	26.34	0.26	19.48
I2SB	1000	36.01	0.07	5.8	30.75	0.2	23.01	25.22	0.26	24.13	26.12	0.27	20.35
DiffPIR	100	28.10	0.13	21.53	31.44	0.10	20.20	20.39	0.36	70.45	-	-	-
DDRM	20	36.73	0.07	4.34	29.21	0.21	19.97	26.05	0.27	46.49	26.33	0.33	47.02

I2SB (Lui, ICML 2023) and CDDb (Chung, NeurIPS 2023) are Schrodinger bridges from y to x . DiffPIR (Zhu, CVPR 2023) and DDRM (Kawar, NeurIPS 2022) are zero-shot DM methods.

ILLUSTRATIVE EXAMPLES

UD2M samples on an MNIST SRx4
problem, for different K & N .



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CONCLUSION

Take-home 1: We construct powerful, explainable and modular image restoration methods by modifying the mathematics of generative AI image models to embed physical models and allow prompting with sensor data.

Take-home 2: The Langevin process and distillation play key roles in our construction - They are central to achieving excellent results with low NFEs.

Take-home 3: Extensions to video restoration through gen. video models.

THANK YOU!

Please feel free to reach out if you'd like to discuss an idea or if you have any questions about our research papers and codes.

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